

LETTER TO THE EDITOR

Modification of the sample size and the schedule of interim analyses in survival trials based on data inspections

by H. Schäfer and H.-H. Müller, *Statistics in Medicine* 2001; **20**:3741–3751

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Schäfer and Müller extend flexible designs using the conditional error probability under the null hypothesis in an unplanned interim look to perform design adaptations for survival trials. They show how to use the independent increment structure of the log rank test statistics under the null hypothesis (based on asymptotic normality) in sequential time to event studies. In flexible designs based on the combination test principle [1] (which is closely related to the conditional error approach) typically stage wise test statistics from stochastically independent samples are combined according to a prefixed combination function. In the application to time to event data, after an interim analysis in general information will be used from patients which have been in the risk set already in the interim analysis, but had no event until then. This structure imposes limitations on the information to be used for adaptations in the interim analysis, which have not been addressed in the paper. In general it is not possible to use all information which is available at the interim analysis for design modifications without compromising the type I error rate even when using adaptive test statistics.

Consider the most extreme case that in a parallel group treatment comparison there is a surrogate variable which perfectly predicts the time of the event. For simplicity assume further that the interim analysis is performed before the first event occurs and that for all recruited patients the surrogate variable is observed. Then the conditional error of the log rank test is equal to α . In this case the procedure of Müller and Schäfer rejects the null hypothesis if the long rank test in the total sample rejects at nominal level α .

The following adaptation strategy leads to an exaggeration of the type I error rate: If the log rank test based on the events to come from the already recruited patients leads to a rejection (which can be predicted from the surrogate variable) do not recruit any further patients. Otherwise take a second chance by increasing the sample size considerably such that the outcome of the patients recruited before the interim analysis has a negligible weight in the log rank test statistics. Since this practically corresponds to two independent level α tests, one for the patients recruited before and the other for those recruited after the interim analysis such a strategy leads to an actual type I error of nearly 2α .

Thus, in order to guarantee the type I error rate without any constraint on the adaptation rule no information correlated with the outcome of patients ‘surviving’ the interim analysis may be used for design adaptations (unless the correlation structure is known and appropriately corrected for). However, any design modifications exclusively based on the log rank test

statistics at the interim analysis and information on patients dying before the interim analysis can be performed safely. Clearly in practical scenarios the anti-conservativeness (if it occurs) will be less pronounced than in the above extreme example. However, care has to be taken when applying flexible designs to time to event data.

This problem is specific for time to event data and does not occur in other settings where the second stage test statistics is computed only with information from patients that have been recruited after the interim analysis or, more generally, where no information on patients entering the second stage test statistics is used for adaptations.

REFERENCE

1. Bauer P, Köhne K. Evaluation of experiments with adaptive interim analyses. *Biometrics* 1994; **50**:1029–1041.

AUTHORS' REPLY

Adaptive designs as first introduced by Bauer and Köhne [1] allow for design modifications during an ongoing clinical trial. Design modifications may include more than just extensions or reductions of the sample size.

In their letter, Bauer and Posch address the following general problem: At an interim analysis, the endpoint of the trial has already been observed for a subset of patients recruited up to the interim analysis, but is yet unknown for the rest of the patients. Bauer and Posch consider the question of making design modifications stochastically depending on yet unobserved endpoints. Stochastic dependency may arise when surrogate endpoints are observed in these patients. Bauer and Posch point at the risk of type I error inflation arising from design modifications depending on future observations. We completely agree with this point. However, this may occur in every trial with prolonged follow-up of patients. It is not specific to time-to-event data. When no information on patients entering the second stage test statistics is used for adaptations, as Bauer and Posch formulate at the end of their letter, the problem will not occur in time-to-event studies either.

The method for sample size modification proposed in our paper is based on the general conditional rejection probability principle for design adaptations introduced in References [2, 3]. In contrast to adaptive methods which rely on independent block structure, the conditional rejection probability principle for design modifications in theory is flexible enough to deal with this problem. The CRP principle simply says that one has to condition on all information which is used for the design adaptation, and to keep the conditional probability of rejecting the null-hypothesis given this information constant under the old and the new trial design. In principle, this makes it possible to use all information which is available at the interim analysis for design modifications without compromising the type I error rate.

In the example given by Bauer and Posch, this principle will safely protect against type I error inflation: when one decides to use a perfect predictor of outcome in future patients for design modifications, as in the example given by Bauer and Posch, one has to condition on this information. Before the first event has occurred, the conditional rejection probability is 1 if the final test leads to rejection and is 0 otherwise.

In practical scenarios, it is of course necessary to restrict design modifications to those informations for which the conditional rejection probability can be determined. In our article,